Department of Electrical and Computer Engineering
North South University

Senior Design Project

EHR-Based Pre-Consultation Medical Documentation System

A Bangla, dialect-aware voice-to-EHR pipeline for pre-consultation patient triage

Submitted by:

Rafiur Rahman Mashrafi	ID# 2221971042
M. G. Rabbi Hossen	ID# 2222516042
Israt Zaman Srity	ID# 2211084042

Faculty Advisor:

Dr. Mohammad Ashrafuzzaman Khan [AzK]

Associate Professor

Department of Electrical and Computer Engineering

Spring, 2026

LETTER OF TRANSMITTAL

April, 2026

To
Dr. Mohammad Abdul Matin [mtn]
Professor & Chair
Department of Electrical and Computer Engineering
North South University, Dhaka

Subject: Submission of Capstone Project Report (CSE499A) on "EHR-Based Pre-Consultation Medical Documentation System."

Dear Sir,

With due respect, we would like to submit our Senior Design I (CSE499A) Capstone Project Report on the "EHR-Based Pre-Consultation Medical Documentation System" as a part of our B.Sc. program in Computer Science and Engineering. The report covers the work completed during the first phase of the project: a multi-dialect Bangla audio corpus (approximately 4.7 hours, five regional varieties), a benchmark of twelve open-source Bangla and multilingual Automatic Speech Recognition (ASR) models on this corpus, and a follow-up benchmark of six larger 1.7B–7B parameter multimodal audio language models on the same data. The report also describes the system design for the full pre-consultation pipeline that will be implemented in CSE499B.

We have tried our best to follow your guidelines, the BAETE Capstone Final Report template, and the requirements set by our faculty advisor. We will be highly obliged if you kindly receive this report and provide your valuable judgment. It would be our immense pleasure if you find this report useful and informative for the assessment of our Senior Design I work.

Sincerely Yours,

.....
Rafiur Rahman Mashrafi
ID# 2221971042

ECE Department
North South University, Bangladesh

.....
M. G. Rabbi Hossen
ID# 2222516042
ECE Department
North South University, Bangladesh

.....
Israt Zaman Srity
ID# 2211084042
ECE Department
North South University, Bangladesh

APPROVAL

Rafiur Rahman Mashrafi (ID# 2221971042), M. G. Rabbi Hossen (ID# 2222516042) and Israt Zaman Srity (ID# 2211084042) from the Department of Electrical and Computer Engineering, North South University, have worked on the Senior Design Project titled "*EHR-Based Pre-Consultation Medical Documentation System*" under the supervision of Dr. Mohammad Ashrafuzzaman Khan [AzK] partial fulfillment of the requirement for the degree of Bachelor of Science in Computer Science and Engineering and have submitted it satisfactorily.

Supervisor's Signature

.....

Dr. Mohammad Ashrafuzzaman Khan [AzK]
Associate Professor
Department of Electrical and Computer Engineering
North South University
Dhaka, Bangladesh.

Chairman's Signature

.....

Dr. Mohammad Abdul Matin [mtn]
Professor & Chair
Ph.D. (Newcastle University, UK)
Department of Electrical and Computer Engineering
North South University
Dhaka, Bangladesh.

DECLARATION

This is to declare that this project is our original work. No part of this work has been submitted elsewhere partially or fully for the award of any other degree or diploma. All project-related information, audio data, model evaluation results, and code have been generated by us during the Spring 2026 semester. Any material taken from other published sources has been duly acknowledged through citations.

Students' names & Signatures

1. Rafiur Rahman Mashrafi

_____

2. M. G. Rabbi Hossen

_____

3. Israt Zaman Srity

_____

ACKNOWLEDGEMENTS

The authors would like to express their heartfelt gratitude towards their project and research supervisor, **Dr. Mohammad Ashrafuzzaman Khan (AzK)**, Associate Professor, Department of Electrical and Computer Engineering, North South University, for his constant guidance, technical mentorship, and constructive feedback throughout the planning and execution of every phase of this Senior Design I project. His emphasis on rigorous evaluation – first on existing Bangla ASR models, then on larger multimodal audio language models – shaped both the methodology and the conclusions of this work.

The authors would also like to thank the **Department of Electrical and Computer Engineering, North South University, Bangladesh**, for facilitating the laboratory access, GPU compute resources, and the academic environment in which this research was carried out.

ABSTRACT

EHR-Based Pre-Consultation Medical Documentation System

This Senior Design I (CSE499A) project investigates the feasibility of a voice-driven, pre-consultation Electronic Health Record (EHR) system for Bangladeshi clinical settings. The proposed pipeline captures a patient's spoken description of symptoms, medications, and history in natural Bangla – including regional dialects and Bangla–English code-mixing – transcribes it through a Bangla-tuned ASR model, extracts clinically relevant entities through a medical Named Entity Recognition (NER) module, and presents a structured EHR with differential-diagnosis hints to the attending physician before the consultation begins. The first phase of the project focused on the hardest sub-problem of the pipeline: speech recognition for real Bangladeshi clinical speech. A multi-dialect Bangla audio corpus of approximately 4.7 hours covering five regional varieties (Puran Dhaka, Barishali, Sylheti, Normal Bangla, and Indian Bangla) was assembled and preprocessed. Twelve open-source Bangla and multilingual ASR models were benchmarked on this corpus, and the best-performing baseline (BengaliAI Regional) achieved a Word Error Rate (WER) of 46.94%, a Character Error Rate (CER) of 24.29%, and a BLEU score of 0.273. A subsequent benchmark of six larger 1.7B–7B parameter multimodal audio language models – including Qwen2-Audio, Qwen3-ASR-1.7B, Voxtral-Mini, Phi-4 Multimodal, and Qwen2.5-Omni – showed that none of these larger models matched the smaller, Bangla-specialized baseline; the closest large model trailed by more than 50 percentage points of WER, and two failed to produce usable output at all. The result establishes that scaling alone does not solve dialect-aware Bangla ASR, and motivates the next phase of the project: parameter-efficient fine-tuning of a mid-sized model (Qwen3-ASR-1.7B) on the collected dialect corpus. All code, evaluation scripts, and the dataset log are publicly available at the project repository on GitHub.

Keywords: Bangla ASR, dialect-aware speech recognition, Electronic Health Records, code-mixed speech, low-resource medical NLP, multimodal audio language models, parameter-efficient fine-tuning.

TABLE OF CONTENTS

Letter of Transmittal **i**

Approval **iii**

Declaration **iv**

Acknowledgements **v**

Abstract **vi**

1 Introduction **1**

1.1 Background and Motivation 1

1.2 Purpose and Goal of the Project 2

1.3 Organization of the Report 3

2 Research Literature Review **4**

2.1 Existing Research and Limitations 4

2.2 Limitations of Existing Work 5

3 Methodology **7**

3.1 System Design 7

3.2 Hardware and Software Components 8

3.3 Hardware and Software Implementation 9

4 Investigation/Experiment, Result, Analysis and Discussion **11**

4.1 Dataset 11

4.2 Phase 2 — Baseline ASR Benchmark 11

4.3 Phase 3 — Larger Multimodal Audio LLM Benchmark 12

4.4 Discussion 13

5 Impacts of the Project **15**

5.1 Impact of this project on societal, health, safety, legal and cultural issues 15

5.2 Impact of this project on environment and sustainability 16

6 Project Planning and Budget **18**

6.1 Project Planning 18

6.2 Project Budget 18

7 Complex Engineering Problems and Activities **20**

7.1 Complex Engineering Problems (CEP) 20

7.2 Complex Engineering Activities (CEA) 22

8 Conclusions **24**

8.1 Summary 24

8.2 Limitations 24

8.3 Future Improvement 25

References **27**

LIST OF FIGURES

3.1 Block diagram of the proposed pre-consultation EHR pipeline. Stage 1 captures patient speech via a tablet/kiosk in the waiting area. Stages 2–4 execute the AI pipeline (ASR → NER → EHR). Stage 5 delivers the structured record to the physician's screen together with the original transcript and differential-diagnosis hints. 7

6.1 Gantt chart of CSE499A execution, March–April 2026. Blue bars denote phase activities; red diamonds mark demonstrations to the faculty advisor; the green bar covers report writing and submission. 18

LIST OF TABLES

3.1 Software tools and components used. 8

4.1 Multi-dialect Bangla audio corpus collected in Phase 1. 11

4.2 Baseline ASR results on the multi-dialect Bangla corpus (Phase 2). Lower WER/CER is better; higher BLEU is better. Best score in each column is in bold. 12

4.3 Larger multimodal audio LLM results on the multi-dialect Bangla corpus (Phase 3), compared against the best baseline. Lower WER/CER is better; higher BLEU is better. 13

6.1 Project budget for CSE499A. 19

7.1 Complex Engineering Problem (CEP) attributes addressed by this project. . . . 20

7.2 Complex Engineering Activity (CEA) attributes addressed by this project. . . . 22

Chapter 1

Introduction

1.1 Background and Motivation

The healthcare delivery system in Bangladesh operates under a chronic imbalance between patient demand and physician supply. According to the World Health Organization, the country has roughly seven physicians per 10,000 population [1], far below the recommended ratio for adequate primary care. Outpatient consultations in public clinics frequently last only a few minutes per patient, with a substantial portion of that limited time consumed by manual history-taking and handwritten note preparation. The shortage of structured Electronic Health Records (EHRs), particularly in small and rural clinics, further compounds the problem: patient histories are not retained, longitudinal continuity of care is lost, and clinically relevant patterns across visits go undocumented.

A second, related gap is technological. Conversational AI, automatic speech recognition, and clinical Natural Language Processing (NLP) have advanced rapidly in English-language healthcare, with commercial "ambient AI scribes" such as Nuance DAX, DeepScribe, and Abridge now deployed in major hospitals abroad. None of these systems support Bangla. The available Bangla speech recognition models have been trained primarily on clean, formal speech (news broadcasts, audiobooks, and read sentences), and our evaluations confirm that their accuracy collapses on real Bangladeshi clinical speech, which is characterized by three properties simultaneously: *regional dialect variation* (Sylheti, Chittagonian, Barishali, Dhakaiya, and others differ in pronunciation, vocabulary, and grammar), *Bangla–English code-mixing* (medical and everyday English terms freely embedded in Bangla sentences), and *informal, unscripted delivery* (hesitations, repetitions, and incomplete utterances). Solving the ASR problem under these three conditions is the foundational prerequisite for any downstream Bangla medical NLP system.

The motivation for this project is therefore both practical and research-oriented. Practically, a working pre-consultation EHR system that handles Bangla dialect speech would relieve documentation burden on doctors, increase patient throughput in overloaded clinics, and introduce digital records to a setting that currently lacks them. Research-wise, evaluating and adapting modern ASR and NLP models for real Bangladeshi clinical speech contributes Bangla-specific

resources and findings that the broader research community can build upon.

1.2 Purpose and Goal of the Project

The purpose of this project is to design, evaluate, and develop a voice-driven pre-consultation EHR system tailored to Bangladeshi clinical practice. The end-to-end pipeline accepts a patient's spoken description in natural Bangla, transcribes it through a dialect-aware ASR model, extracts clinically relevant entities (symptoms, medications, dosage and duration, allergies, and prior conditions) through a medical NER module, structures the extracted entities into a standardized EHR record, and presents it to the attending physician along with a short list of differential-diagnosis hints derived from a lightweight medical knowledge layer. The system is explicitly an *assistive* tool: it does not issue diagnoses; the final clinical decision remains with the licensed physician, and the original transcript is always presented alongside the extracted EHR for verification.

The specific goals of CSE499A (Senior Design I) – the scope of the present report – are:

1. To collect a multi-dialect Bangla audio corpus that reflects the actual speech characteristics of Bangladeshi patients;
2. To benchmark existing open-source Bangla and multilingual ASR models on this corpus and identify the strongest baseline;
3. To evaluate whether larger 2B–7B parameter multimodal audio language models improve performance on dialect Bangla speech; and
4. To select a target model and methodology for the fine-tuning and downstream NLP work to be carried out in CSE499B.

The novelty of the project lies in the explicit focus on *dialect-aware, code-mixed Bangla clinical speech* – an area that, to the best of our knowledge, has not been benchmarked in the existing literature – and in the empirical finding that scale alone (through 2B–7B parameter audio language models) does not solve this problem.

1.3 Organization of the Report

The remainder of this report is organized as follows. Chapter 2 presents a review of the related research in multilingual ASR, Bengali speech resources, biomedical and clinical NLP, and AI clinical scribes, and identifies the limitations that motivate this project. Chapter 3 describes the overall system design, the software components employed, and the implementation methodology used in the model evaluations. Chapter 4 presents the experiments, results, and analysis of the two model benchmarks (twelve baseline models, and six larger multimodal audio LLMs). Chapter 5 discusses the societal, health, legal, and environmental impacts of the project. Chapter 6 presents the project planning timeline through a Gantt chart and the project budget. Chapter 7 maps the work to the Complex Engineering Problem (CEP) and Complex Engineering Activity (CEA) attributes required for BAETE accreditation. Chapter 8 concludes the report with a summary, the limitations of the work completed in CSE499A, and the planned improvements for CSE499B.

Chapter 2

Research Literature Review

2.1 Existing Research and Limitations

Multilingual and large-scale ASR

Modern automatic speech recognition is dominated by large self-supervised and supervised neural models trained on web-scale multilingual audio. Radford *et al.* [2] introduced **Whisper**, an encoder–decoder Transformer trained on 680,000 hours of weakly-labelled multilingual audio that achieves strong zero-shot performance across nearly 100 languages, including Bangla. Pratap *et al.* [3] released the **Massively Multilingual Speech** (MMS) project, which extends ASR coverage to over 1,100 languages by combining religious-text audio with self-supervised pre-training and CTC fine-tuning. Baevski *et al.* [4] proposed **wav2vec 2.0**, a self-supervised speech representation framework that underpins many downstream Bangla ASR systems via cross-lingual transfer (XLSR-53, XLSR-300M). The Seamless Communication team's **SeamlessM4T** [5] unifies speech-to-text, text-to-text, and speech-to-speech translation across a wide range of languages.

More recent work has shifted toward *multimodal audio language models* that fuse speech encoders with autoregressive language model decoders. Chu *et al.* [6] introduced the Qwen2-Audio family, and the same authors followed with the specialized Qwen3-ASR-1.7B model. Mistral AI's Voxtral-Mini and Microsoft's Phi-4 Multimodal extend this trend. While these models demonstrate impressive multilingual transcription on benchmark corpora, their performance on out-of-distribution low-resource language settings – particularly dialect speech – has been studied relatively little.

Bengali speech resources

For Bengali specifically, several public resources exist. The **Bengali.AI Speech Recognition Competition** [7] on Kaggle in 2023 produced a benchmark of read-aloud Bengali speech and motivated several fine-tuned variants of Whisper and Wav2Vec 2.0. The **Mozilla Common Voice** [8] project provides a crowdsourced Bengali speech corpus collected from volunteers

worldwide. AI4Bharat's **IndicWav2Vec** [9] extends self-supervised speech models to Indic languages, and the **Vakyansh** [10] project provides ASR models tuned for Indic ASR including Bengali. Most relevantly to this project, Rakib *et al.* [11] released the **OOD-Speech** dataset, an out-of-distribution Bengali ASR corpus that explicitly tests model robustness on speech that differs from the training distribution; this is the closest published work to the dialect-aware focus of our project.

Biomedical and clinical NLP

On the language-modeling side, Bhattacharjee *et al.* [12] introduced **BanglaBERT**, a Bangla BERT-style language model with benchmarks for multiple downstream Bangla NLP tasks; this provides a strong starting point for any future Bangla medical NER work in the project. For the medical domain, Lee *et al.* [13] released **BioBERT**, a BERT model pretrained on PubMed abstracts and PMC full-text articles, and Alsentzer *et al.* [14] released **ClinicalBERT**, pretrained on MIMIC-III clinical notes. Both established that domain-adaptive pretraining substantially improves performance on downstream clinical extraction tasks. The original Transformer architecture [15] and the BERT encoder [16] underpin all of the above. The Hugging Face Transformers library [17] provides reference implementations of these models and is the primary tooling used in this project's evaluations.

Parameter-efficient fine-tuning

Hu *et al.* [18] introduced **Low-Rank Adaptation (LoRA)**, which freezes the base model weights and trains only small low-rank adapter matrices, dramatically reducing the compute and memory footprint of fine-tuning. LoRA and related parameter-efficient fine-tuning (PEFT) techniques are now the standard approach for adapting large speech and language models to low-resource domains, and are the planned methodology for the CSE499B fine-tuning phase of this project. Bain *et al.*'s **WhisperX** [19] provides robust long-form transcription with forced alignment and is used in the audio preprocessing pipeline.

2.2 Limitations of Existing Work

The following limitations have been identified after reviewing the related literature, motivating the present project:

1. **No published benchmarks on dialect-aware Bangladeshi clinical speech.** Existing Bangla ASR benchmarks are dominated by clean, read-aloud speech (Common Voice, BengaliAI competition data, read news). The closest exception, OOD-Speech [11], explicitly tests out-of-distribution Bengali but does not focus on the simultaneous combination of dialect, code-mixing, and clinical/informal register that is characteristic of real patient speech in Bangladeshi clinics.

2. **Code-mixing is under-served.** Most Bengali ASR work either explicitly cleans out English tokens or treats the language as monolingual. In practice, Bangladeshi speech in medical settings is heavily code-mixed.

3. **No Bangla medical NER benchmark exists.** Biomedical and clinical NER work [13, 14] is overwhelmingly English-only. There is no publicly available Bangla medical entity recognition dataset, and no published Bangla equivalent of BioBERT or ClinicalBERT.

4. **Behavior of large multimodal audio LLMs on low-resource dialect speech is largely unstudied.** The recent Qwen2-Audio, Qwen3-ASR, Phi-4 Multimodal, Voxtral, and Qwen2.5-Omni families [6] report strong performance on high-resource language benchmarks but do not systematically report performance on Bangla dialect speech. The empirical investigation in Chapter 4 addresses this gap directly.

5. **No end-to-end Bangla pre-consultation EHR system has been published.** Existing AI clinical scribes (Nuance DAX, DeepScribe, Abridge) target English-language deployments in Western hospitals. There is no comparable Bangla-language system, and the Bangladesh primary-care setting has additional constraints (intermittent connectivity, cost sensitivity, low patient digital literacy) that demand a fundamentally different design.

Chapter 3

Methodology

3.1 System Design

The proposed system is a five-stage pipeline that converts a patient's spoken description into a structured EHR record presented to the attending physician. Figure 3.1 shows the overall block diagram.

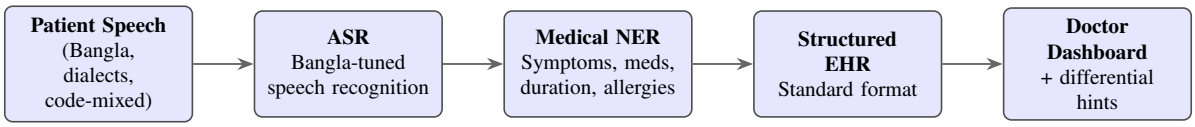

Figure 3.1: Block diagram of the proposed pre-consultation EHR pipeline. Stage 1 captures patient speech via a tablet/kiosk in the waiting area. Stages 2–4 execute the AI pipeline (ASR $\rightarrow$ NER $\rightarrow$ EHR). Stage 5 delivers the structured record to the physician's screen together with the original transcript and differential-diagnosis hints.

Stage descriptions

Stage 1 – Patient Speech. The patient sits at a voice-enabled device (tablet or kiosk) in the clinic waiting area. After a recorded informed-consent prompt (in Bangla), the patient describes their symptoms, medical history, current medications, and allergies in natural speech. No structured questionnaire is imposed; the patient speaks freely.

Stage 2 – ASR. The recorded audio is passed to a Bangla-tuned ASR model that produces a Unicode Bangla transcript with embedded English words preserved as-is (code-mixing is not normalized out). This is the focus of CSE499A, and is described in detail in Chapter 4.

Stage 3 – Medical NER. A medical Named Entity Recognition module reads the transcript and labels clinically relevant spans: *symptom*, *disease*, *medication*, *dosage*, *duration*, *frequency*, *allergy*, and *anatomical site*. The planned implementation will adapt BanglaBERT [12] with a token classification head, trained on a small annotated corpus to be collected during CSE499B.

Stage 4 – Structured EHR. The labelled spans are normalized (e.g., dosage strings to standard units, durations to ISO 8601-like intervals) and packaged into a structured record.

The schema includes patient identifier, recording timestamp, chief complaint, list of symptoms with onset/duration, medications, allergies, and a free-text notes field containing the verbatim transcript. The schema design is the deliverable of Phase 5, scheduled for the next demo.

Stage 5 – Doctor Dashboard. The physician sees a clean web interface displaying the structured record, the original transcript (for verification), and a short list of differential-diagnosis hints derived from a lightweight medical knowledge graph. The system never renders a diagnosis; it only narrows the search space.

3.2 Hardware and Software Components

This is a software-only project; no custom hardware has been fabricated. The patient-facing capture device, in deployment, would be a consumer tablet or kiosk running the recording front-end. All model evaluation and development was performed on free-tier Google Colab GPU instances. Table 3.1 lists the principal software tools and their justification.

Table 3.1: Software tools and components used.

Tool	**Functions**	**Other similar tools (if any)**	**Why selected this tool**
Python 3.10	Project implementation language	R, Julia	Standard for ML/NLP; widest ecosystem of speech/NLP libraries.
PyTorch 2.x	Deep learning framework	TensorFlow, JAX	Native support across all evaluated ASR models; preferred by Hugging Face.
Hugging Face Transformers [17]	Model loading, inference, tokenization	ESPnet, NeMo	Single unified API across Whisper, Wav2Vec2, MMS, SeamlessM4T, Qwen-Audio, Voxtral, Phi-4.
HF Datasets, Accelerate, PEFT	Dataset streaming, mixed-precision inference, planned LoRA fine-tuning	Custom data pipelines, DeepSpeed	Lowest friction with Transformers; PEFT [18] is the planned fine-tuning route.
Librosa, SoundFile, FFmpeg	Audio I/O, resampling to 16 kHz, mono conversion	torchaudio, sox	Stable, well-documented, run reliably on Colab.

continued on next page

Table 3.1 continued from previous page

Tool	Functions	Other similar tools (if any)	Why selected this tool
WhisperX [19]	Long-form audio segmentation and forced alignment	pyannote.audio	Required for transcribing audio files longer than Whisper's 30-second window.
`jiwer`, `evaluate` (HF)	Computing WER, CER, BLEU	sclite (NIST)	Pure-Python; no external binary dependency; standard for speech evaluation.
yt-dlp	Audio collection from public sources	youtube-dl	Maintained fork; reliable for our dialect audio sourcing.
Google Colab (T4, A100)	Primary GPU compute	Kaggle, local GPU	Free tier sufficient for inference benchmarks.
Google Drive	Dataset and checkpoint storage	S3, local NAS	Free integration with Colab; sufficient for ∼5 GB project data.
Git / GitHub	Source control and project repository	GitLab, Bitbucket	Standard; required for reproducibility and submission.

3.3 Hardware and Software Implementation

The implementation work in CSE499A spanned four operational phases, each closed by a demonstration to the faculty advisor. The flow is summarized below; detailed results appear in Chapter 4.

Phase 1 – Audio collection and preprocessing. Audio in five regional Bangla varieties (Puran Dhaka, Barishali, Sylheti, Normal Bangla, and Indian Bangla) was sourced from publicly available recordings and organized under a fixed naming convention of the form `[dialect]_[index]_[gender]_[age].mp3`. Each file's metadata was logged in `dataset_log.csv`: filename, source, dialect code, speaker gender, approximate age, and ffprobe-measured duration in seconds. The corpus was then preprocessed: resampled to 16 kHz mono PCM, RMS-normalized, and segmented into 10–15 second clips using WhisperX's voice-activity detector. The segmentation step is necessary because most evaluated ASR models cannot accept long-form input directly.

Phase 2 – Baseline ASR benchmark. Twelve open-source Bangla and multilingual ASR models were evaluated on the segmented corpus. For each model, the entire corpus was transcribed automatically; the resulting hypothesis transcripts were scored against reference transcripts using three standard metrics: Word Error Rate (WER), Character Error Rate (CER), and corpus-level BLEU. WER and CER are computed by the `jiwer` library after light reference–hypothesis normalization (whitespace, punctuation, Unicode normalization to NFC form); BLEU is computed by the Hugging Face `evaluate` library. The full evaluation table is provided in the project repository.

Phase 3 – Larger multimodal audio LLM benchmark. Six larger 1.7B–7B parameter multimodal audio language models were evaluated on the same corpus, using the same metrics, to test whether scaling alone closes the dialect ASR gap. Two models (Phi-4 Multimodal and Qwen2.5-Omni) failed to produce usable output and are recorded as such.

Phase 4 – Fine-tuning research (in progress). Following advisor guidance, the team is currently researching the methodology for parameter-efficient fine-tuning of the **Qwen3-ASR-1.7B** model on the collected dialect corpus. The plan, scheduled for execution in CSE499B, will use LoRA [18] adapters on the speech encoder while keeping the language-model decoder weights frozen. No training has been performed yet; this phase is currently a methodology study and is expected to be completed in CSE499B.

Phase 5 – Comparative chatbot output study (selected, not yet executed). As a parallel deliverable, the team has been asked to prompt several public AI chatbots (ChatGPT, DeepSeek, Qwen, Perplexity) with simulated patient interactions and to compare how each chatbot renders the resulting EHR-style structured output. This will inform the schema design for Stage 4 of the pipeline. This phase is scheduled for the next demo and is not part of CSE499A's results chapter.

Chapter 4

Investigation/Experiment, Result, Analysis and Discussion

This chapter reports the experimental results from the two ASR evaluations conducted during CSE499A: the twelve-model baseline benchmark (Phase 2) and the six-model larger multimodal audio LLM benchmark (Phase 3). All results are computed on the same multi-dialect corpus described below.

4.1 Dataset

The corpus assembled for evaluation consists of approximately 4.7 hours of Bangla audio across five regional varieties. Table 4.1 summarizes the per-dialect counts and durations. The duration figures are computed directly from the audio files using `ffprobe`.

Table 4.1: Multi-dialect Bangla audio corpus collected in Phase 1.

Dialect	**Code**	**Files**	**Duration**
Puran Dhaka (Dhakaiya)	pd	1	35.7 min
Barishali	br	2	33.8 min
Sylheti	sy	2	31.7 min
Normal Bangla	nb	3	87.0 min
Indian Bangla	ib	2	95.9 min
Total		**10**	**284.1 min ($\approx$ 4.7 hr)**

The audio is gender-balanced where source availability permitted (8 male, 2 female speakers across 10 files), and ages span the 20s–50s range. The recordings include both monologue and dialogue formats, contain natural code-mixing (English medical and everyday terms embedded in Bangla sentences), and are sourced from publicly available recordings; no clinical patient data has been used. After segmentation into 10–15 second clips at the voice-activity boundaries identified by WhisperX, the corpus yields approximately 1,100 evaluation segments.

4.2 Phase 2 — Baseline ASR Benchmark

Twelve open-source Bangla and multilingual ASR models were evaluated on the corpus. Table 4.2 reports WER, CER, and BLEU for each, sorted in ascending WER (lower is better for WER and CER; higher is better for BLEU).

Table 4.2: Baseline ASR results on the multi-dialect Bangla corpus (Phase 2). Lower WER/CER is better; higher BLEU is better. Best score in each column is in bold.

Model	**WER ↓**	**CER ↓**	**BLEU ↑**
BengaliAI Regional	**0.4694**	**0.2429**	**0.2730**
BengaliAI Whisper	0.5055	0.3075	0.2527
MMS [3]	0.5568	0.2687	0.2003
Vakyansh-Bn [10]	0.6142	0.2950	0.1810
Wav2Vec2-XLSR-CV-Bn [4]	0.6808	0.4365	0.1668
Wav2Vec2-XLSR-300M-Bn [4]	0.7023	0.3894	0.1191
SeamlessM4T [5]	0.7679	0.5210	0.1233
Whisper Large-v3 [2]	0.9097	0.5827	0.0646
Wav2Vec2-XLSR-53-Bn [4]	1.0023	0.8245	0.0002
Whisper-v3 Turbo [2]	1.0264	0.7850	0.0117
Whisper Medium [2]	1.1680	1.2299	0.0002
Whisper Small [2]	1.9580	1.7060	0.0000

Analysis

The Bangla-specialized smaller models occupy the top of the leaderboard. BengaliAI Regional achieves the best result on every metric (WER 46.94%, CER 24.29%, BLEU 0.273), followed closely by BengaliAI Whisper, MMS, and Vakyansh-Bn. Multilingual models that are not specifically tuned for Bangla – particularly the Whisper family beyond Whisper Large-v3 – perform substantially worse, with Whisper Small reaching a WER of 195.8% (a WER above 100% indicates that the hypothesis contains more substitutions and insertions combined than there are reference words, i.e. the model is hallucinating). SeamlessM4T's CER is significantly higher than its WER would suggest, which indicates frequent character-level errors within otherwise acceptable word boundaries.

The principal observation from this table is that even the best baseline gets nearly half the words wrong. Bangla dialect ASR is not a solved problem with off-the-shelf models trained on news, audiobooks, or read sentences. This finding is consistent with the out-of-distribution behaviour reported by Rakib *et al.* [11] for OOD-Speech, and it directly motivates the next experiment.

4.3 Phase 3 — Larger Multimodal Audio LLM Benchmark

Following the advisor's guidance, six larger 1.7B–7B parameter multimodal audio language models were evaluated on the same corpus to test the hypothesis that scaling alone improves dialect Bangla ASR. Table 4.3 reports the results.

Table 4.3: Larger multimodal audio LLM results on the multi-dialect Bangla corpus (Phase 3), compared against the best baseline. Lower WER/CER is better; higher BLEU is better.

Model	**Params**	**WER ↓**	**CER ↓**	**BLEU ↑**
BengaliAI Regional (best baseline)	–	**0.4694**	**0.2429**	**0.2730**
Qwen2-Audio (via English) [6]	7B	1.0039	0.9214	0.0001
Qwen2-Audio (direct Bangla) [6]	7B	1.0837	0.9327	0.0003
Qwen3-ASR-1.7B [6]	1.7B	1.1339	0.8962	0.0031
Voxtral-Mini	3B	1.3627	0.9722	0.0626
Phi-4 Multimodal	5.6B	–	–	–
Qwen2.5-Omni	7B	–	–	–

Analysis – Key Finding

Every single one of the larger 1.7B–7B parameter audio language models performed substantially worse than the smaller, Bangla-tuned baseline. The gap is large – the closest large model (Qwen2-Audio via English routing) trails BengaliAI Regional by **53.5 percentage points of WER**. Two models (Phi-4 Multimodal and Qwen2.5-Omni) failed entirely: the former emitted empty or garbled output; the latter routed the input through its unified omni-modal pipeline in a way that did not yield a Bengali transcript.

The headline finding from CSE499A is therefore:

> *Bigger is not better for Bangla dialect speech.* A specialized smaller model outperforms general-purpose 1.7B–7B parameter multimodal audio LLMs by a margin of more than 50 WER points. Scale alone does not transfer to dialect-aware Bangladeshi clinical speech.

Three failure modes were observed in the larger models on manual inspection of the transcripts:

1. **Over-translation.** Audio LLMs trained primarily on high-resource languages tend to translate, paraphrase, or summarize the input rather than transcribing it verbatim. This was particularly evident in Qwen2-Audio (direct Bangla mode).
2. **Hallucination.** The models occasionally generated plausible-sounding Bangla text that was unrelated to the audio, particularly on segments containing dialect or code-mixing they did not handle.
3. **Mis-segmentation.** The long-form input was occasionally chunked incorrectly by the model's internal audio handler, producing either truncated or duplicated transcripts.

4.4 Discussion

The empirical results from Phases 2 and 3 agree on a single conclusion: addressing dialect-aware Bangla clinical ASR will require *specialization*, not raw scale. The best off-the-shelf result (BengaliAI Regional, 46.94% WER) is too inaccurate for clinical deployment, where the downstream NER module would compound any ASR error rate.

Closing the gap will require fine-tuning a model on the kind of speech the system is meant to serve. The choice of target model for fine-tuning is informed by these results: a mid-sized model with a strong Bangla starting point and a fine-tuning-friendly architecture is preferable to either a small purpose-built model (limited capacity) or a 7B-parameter multimodal LLM (poor starting point on dialect Bangla, expensive to adapt). **Qwen3-ASR-1.7B** sits in this sweet spot and has been selected as the target model for the fine-tuning study planned in CSE499B.

Chapter 5

Impacts of the Project

5.1 Impact of this project on societal, health, safety, legal and cultural issues

Societal impact

The project addresses three persistent gaps in Bangladeshi primary healthcare simultaneously: the absence of digital records in most clinics, the chronic shortage of consultation time per patient, and the absence of usable Bangla medical AI tooling. By absorbing the documentation burden before the consultation begins, the system allows physicians to dedicate scarce consultation minutes to examination and clinical reasoning rather than to typing or handwriting notes. The voice-first interface is specifically designed to reach patients who are not literate enough to type or to use English-language health apps – the majority of patients outside metropolitan Dhaka. In doing so, the project narrows the urban–rural digital health divide rather than widening it, which is a common failure mode of healthcare-IT introductions in low-resource settings.

Health impact

The downstream effects on patient health are indirect but real. Structured EHR records enable longitudinal patient care – the physician at follow-up visits can review the prior structured history rather than relying on patient recall. The differential-diagnosis hint module provides a gentle prompt that flags conditions worth ruling out (for example, a persistent cough combined with night sweats and weight loss flags possible tuberculosis). Crucially, the system never issues a diagnosis. It narrows the search space and reminds the physician of possibilities they may want to investigate. The final clinical decision rests with the licensed physician, and the original transcript is always displayed alongside the extracted entities so that the physician can verify the system's understanding.

Safety impact

Three safety-related design properties were enforced in the pipeline. First, the system is *auditable*: every extracted entity in the EHR can be traced back to the audio segment it came from, eliminating silent edits or untraceable model "corrections." Second, the system reports *honest accuracy metrics* – the WER, CER, and BLEU figures reported in Chapter 4 are computed on real dialect audio, not on clean read-aloud benchmarks, so the physician sees an honest view of the system's reliability. Third, the system respects the boundary between *decision support* and *decision making*: no prescription, dosage, or diagnosis is ever auto-confirmed.

Legal impact

Patient health information in Bangladesh is sensitive personal data and falls under the country's emerging data-protection framework. The deployed version of the system will require: (i) explicit recorded informed consent in the patient's own language before any audio capture; (ii) encryption of audio and EHR records both at rest and in transit; (iii) role-based access control on the doctor-facing dashboard; and (iv) the option for patients to request deletion of their records. Where feasible, the deployed infrastructure will keep sensitive health data on local servers rather than on foreign cloud providers, both for data-sovereignty reasons and to reduce the attack surface for external breach.

Cultural impact

The project is, at its core, an act of *Bangla-language preservation* in technical AI tooling. By explicitly collecting, modelling, and reporting on regional varieties (Sylheti, Chittagonian, Barishali, Dhakaiya) rather than collapsing all input to a single "standard Bangla," the project legitimizes these dialects in the AI infrastructure that increasingly mediates daily life. In the medical context, this has the additional cultural benefit of allowing patients to describe their symptoms in their own dialect rather than being forced to "correct" their speech to match the system.

5.2 Impact of this project on environment and sustainability

The project's environmental footprint is dominated by GPU electricity consumption during model training and inference. The following design choices reduce this footprint:

1. **Inference-only evaluation in CSE499A.** All twelve baseline and six larger models were used in inference mode. No model was trained from scratch, and no end-to-end training run was conducted. The total GPU time consumed in CSE499A is therefore on the order of tens of GPU-hours (single T4/A100 instances), not GPU-weeks.
2. **Selection of a mid-sized model for fine-tuning.** The decision to fine-tune Qwen3-ASR-1.7B rather than a 7B+ parameter audio LLM is justified by the empirical results in Chapter 4 but also has a sustainability rationale: the smaller model trains faster, infers faster at deployment, and requires fewer GPU resources both in development and in production.
3. **Parameter-efficient fine-tuning.** The planned fine-tuning approach will use LoRA [18] adapters, which train only a small fraction of the full model's parameters. This reduces fine-tuning compute by approximately one to two orders of magnitude versus full fine-tuning, with proportional reduction in energy consumption.
4. **Edge inference at deployment.** A small clinic kiosk running a fine-tuned 1.7B model can be served on a single mid-range GPU or even a quantized CPU build, avoiding the need for continuous high-power cloud inference for every patient interaction.

The project supports the **UN Sustainable Development Goals #3 (Good Health and Well-being)**, **#4 (Quality Education – through resource creation for Bangla NLP)**, **#9 (Industry, Innovation and Infrastructure)**, and **#10 (Reduced Inequalities – by serving rural and dialect-speaking populations under-served by existing English-only digital health tools)**.

Chapter 6

Project Planning and Budget

6.1 Project Planning

The CSE499A project was structured around five phases delivered through periodic demonstrations to the faculty advisor. The Gantt chart in Figure 6.1 shows the actual execution timeline, spanning the second half of Spring 2026 (March–April 2026).

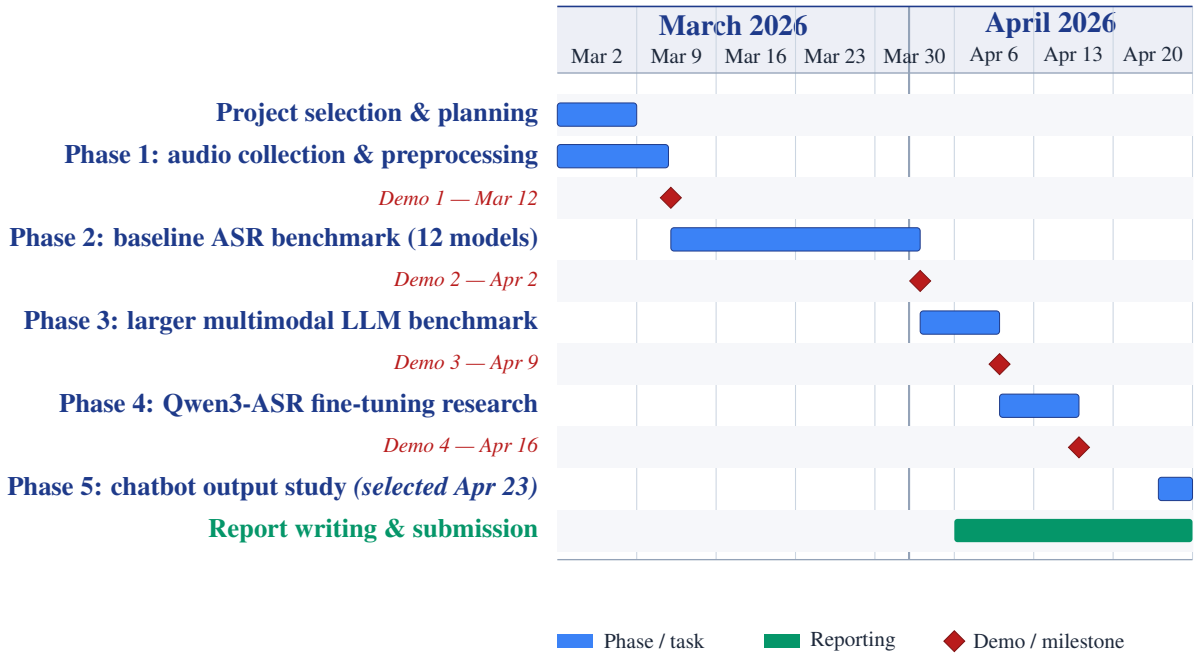

Figure 6.1: Gantt chart of CSE499A execution, March–April 2026. Blue bars denote phase activities; red diamonds mark demonstrations to the faculty advisor; the green bar covers report writing and submission.

The phases listed in Figure 6.1 correspond to the implementation activities described in Chapter 3. Phases 1, 2, and 3 are complete; Phase 4 is in progress as a methodology study; Phase 5 has been selected and will be the next deliverable.

6.2 Project Budget

The CSE499A phase of the project incurred **zero monetary cost** (BDT 0). All compute was performed on the free tier of Google Colab, all evaluated models were obtained from public Hugging Face repositories under their respective open-source licenses, all audio data was sourced from publicly available recordings, and code was hosted free of charge on GitHub. Table 6.1 itemises the resource utilization.

Table 6.1: Project budget for CSE499A.

Item	**Source / Justification**	**Cost (BDT)**
GPU compute (T4 / A100, ∼60 GPU-hours)	Google Colab free tier	0
ASR model checkpoints (12 baseline + 6 larger)	Hugging Face Hub, open licenses	0
Audio dataset (∼4.7 hours, 5 dialects)	Public sources, manual collection	0
Storage (∼5 GB project data)	Google Drive free tier (15 GB)	0
Source-code hosting	GitHub free tier	0
Software (Python, PyTorch, Transformers, etc.)	Open-source	0
Total cost (CSE499A)		**0**

Chapter 7

Complex Engineering Problems and Activities

7.1 Complex Engineering Problems (CEP)

This project meets the BAETE Complex Engineering Problem (CEP) attributes as detailed in Table 7.1. Mappings of the required knowledge depths (K3–K8) reflect the software-, machine-learning-, and signal-processing-oriented nature of the work and have been discussed with the faculty advisor.

Table 7.1: Complex Engineering Problem (CEP) attributes addressed by this project.

	Attributes	Addressing the complex engineering problems (P) in the project
P1	Depth of knowledge required (K3–K8)	The project requires depth in software engineering and data structures (K3), specialist knowledge of machine learning, deep learning, signal processing for ASR, and natural language processing (K4), system architecture and pipeline design for an end-to-end voice-to-EHR system (K5), competence in modern engineering and IT tools including Python, PyTorch, Hugging Face Transformers, Librosa, and FFmpeg (K6), comprehension of the role of engineering in society including healthcare ethics, patient privacy, and responsible AI in clinical settings (K7), and engagement with current peer-reviewed research literature on multilingual ASR, biomedical NLP, and parameter-efficient fine-tuning (K8).
P2	Range of conflicting requirements	Trade-offs exist between transcription accuracy and inference latency on deployment hardware; between model size (and therefore deployment cost) and transcription quality; between strict patient-data confidentiality and the need to log misrecognitions for model improvement; and between dialect coverage and per-dialect accuracy under a fixed data budget.

continued on next page

Table 7.1 continued from previous page

	Attributes	Addressing the complex engineering problems (P) in the project
P3	Depth of analysis required	There is no unique correct design. Selecting the target ASR model requires comparative empirical analysis across twelve baseline models and six larger multimodal LLMs, evaluation under three metrics (WER, CER, BLEU), and reasoned selection considering both accuracy and fine-tuning feasibility. The fine-tuning methodology itself (full vs. LoRA vs. prefix-tuning) requires further analysis.
P4	Familiarity of issues	The team has acquired familiarity with diverse open-source ASR families (Whisper, Wav2Vec 2.0, MMS, SeamlessM4T, BengaliAI, Vakyansh, Qwen2/3-Audio, Phi-4 Multimodal, Voxtral, Qwen2.5-Omni), audio preprocessing pipelines (resampling, VAD-based segmentation, forced alignment with WhisperX), and standard speech-evaluation metrics.
P5	Extent of applicable codes	There is no single applicable engineering code or standard governing this work. The project follows industry conventions for software engineering (semantic versioning, code review through Git), the *IEEE Code of Ethics*, and the *IEEE/ACM Code of Ethics and Professional Conduct* for software engineering. The deployed system will additionally need to comply with the emerging Bangladesh data-protection framework.
P6	Extent of stakeholder involvement	Multiple stakeholders are involved or affected: patients (the data subjects and end-users at the kiosk), physicians (the consumers of the structured EHR), clinic administrators (deployment owners), the Department of Electrical and Computer Engineering at North South University (academic stakeholder), the faculty advisor (technical oversight), and ultimately the relevant national health-data regulators.

continued on next page

Table 7.1 continued from previous page

	Attributes	Addressing the complex engineering problems (P) in the project
P7	Interdependence	The project consists of multiple interdependent sub-systems that must operate correctly in series: audio capture, ASR, medical NER, EHR schema mapping, differential-diagnosis hint generation, and the doctor-facing dashboard. An error or failure in any earlier stage propagates to later stages, requiring careful interface design and per-stage evaluation.

7.2 Complex Engineering Activities (CEA)

This project meets the BAETE Complex Engineering Activity (CEA) attributes as detailed in Table 7.2.

Table 7.2: Complex Engineering Activity (CEA) attributes addressed by this project.

	Attributes	Addressing the complex engineering activities (A) in the project
A1	Range of resources	The project draws on human resources (the three-member project team plus faculty advisor), modern software tools (Python, PyTorch, Hugging Face Transformers, WhisperX, Librosa, FFmpeg), open-source model checkpoints from public model hubs, free-tier cloud GPU compute (Google Colab, T4/A100 instances), peer-reviewed research literature, and a self-assembled multi-dialect Bangla audio corpus.
A2	Level of interactions	The activity requires interaction between project team members (weekly working sessions), with the faculty advisor (periodic demos and design reviews), and indirectly with the open-source machine-learning community (consulting model documentation, open issues on Hugging Face Hub, and published evaluation methodologies). The deployed system will further interact with clinics, physicians, patients, and health-data regulators.

continued on next page

Table 7.2 continued from previous page

	Attributes	Addressing the complex engineering activities (A) in the project
A3	Innovation	The project applies modern multilingual ASR and parameter-efficient fine-tuning techniques to a setting – dialect-aware Bangladeshi clinical speech – that has not been benchmarked in the published literature. The empirical finding that 1.7B–7B parameter multimodal audio LLMs underperform a smaller Bangla-specialized baseline by more than 50 WER points on this corpus is, to the best of the team's knowledge, a novel result.
A4	Consequences to society / Environment	Successful deployment would reduce documentation burden on overworked Bangladeshi physicians, introduce structured digital records to clinics that currently have none, and serve patients under-served by English-only digital health tools (rural populations and dialect speakers). The environmental footprint is mitigated by inference-only evaluation in CSE499A and a planned parameter-efficient fine-tuning approach for CSE499B. The project contributes to UN SDG #3 (Good Health and Well-being) and SDG #10 (Reduced Inequalities).
A5	Familiarity	The activity requires familiarity with deep learning frameworks (PyTorch), the Transformers library, the Whisper / Wav2Vec 2.0 / MMS / Qwen-Audio model families, audio signal processing (resampling, VAD, MFCC and log-Mel front-ends), low-resource language NLP, biomedical NLP literature, and software engineering practice including version control, reproducible evaluation, and documentation.

Chapter 8

Conclusions

8.1 Summary

This Senior Design I (CSE499A) report has presented the first phase of the EHR-Based Pre-Consultation Medical Documentation System project. The proposed end-to-end pipeline – Bangla speech → ASR → medical NER → structured EHR → doctor dashboard – was specified in detail, and the foundational sub-problem (Bangla dialect ASR) was empirically investigated. A multi-dialect Bangla audio corpus of approximately 4.7 hours covering five regional varieties was assembled and preprocessed. Twelve open-source Bangla and multilingual ASR models were evaluated on this corpus, with BengaliAI Regional emerging as the strongest baseline at 46.94% WER, 24.29% CER, and 0.273 BLEU. Six larger 1.7B–7B parameter multimodal audio language models were then evaluated on the same corpus, and *none of them matched the smaller specialized baseline*; the closest large model trailed by more than 50 percentage points of WER, and two failed entirely. These results establish that scaling alone does not solve dialect-aware Bangla ASR and motivate a parameter-efficient fine-tuning approach (target model: Qwen3-ASR-1.7B) for the remainder of the project. All code, evaluation scripts, and the dataset log are publicly available at the project's GitHub repository.

8.2 Limitations

The CSE499A work has the following limitations, all of which the team is explicit about:

1. **Small evaluation corpus.** The 4.7-hour, 10-file corpus is sufficient for an indicative evaluation across five dialects but is too small for robust per-dialect statistical comparison. A larger corpus is needed to confidently distinguish, for example, whether a given model's performance differs significantly between Sylheti and Barishali speakers.
2. **Dialect coverage is incomplete.** The five varieties covered (Puran Dhaka, Barishali, Sylheti, Normal Bangla, Indian Bangla) leave out important Bangladeshi dialects, most notably Chittagonian and Rangpuri. The conclusions reported here apply to the five dialects evaluated and should not be extrapolated to dialects outside this set.
3. **Audio source domain.** The audio used in this evaluation is sourced from publicly available recordings rather than from real clinical encounters. Real clinic audio includes additional challenges (background noise, partial overlapping speech, soft- spoken patients) that have not been tested in CSE499A.
4. **ASR-only evaluation.** Of the five pipeline stages (audio capture, ASR, medical NER, EHR schema, doctor dashboard), only the ASR stage has been evaluated empirically. The medical NER module, the EHR schema, the differential-diagnosis layer, and the doctor-facing dashboard have been specified in design but not yet implemented.

5. **No fine-tuning has been performed.** Phase 4 is at the methodology-research stage. No training runs have been executed. The selection of Qwen3-ASR-1.7B as the target model is based on the benchmark results plus its mid-sized parameter count and fine-tuning-friendly architecture, but the empirical case for the selection is not yet established.
6. **Reference transcripts are partial.** Reference transcripts for the evaluation corpus were prepared by the team, not by certified clinical transcribers, and are subject to transcriber error. The reported WER, CER, and BLEU values are therefore lower bounds on the true error rates.
7. **No deployment study.** No usability testing, latency benchmarking, or clinical-workflow integration study has been conducted. The deployment-side viability of the pipeline (kiosk form factor, doctor-dashboard usability) remains theoretical.

8.3 Future Improvement

CSE499B will address the limitations above through the following planned improvements:

1. **Phase 5 – Comparative chatbot output study (next demo).** Patient-style queries will be presented to ChatGPT, DeepSeek, Qwen, and Perplexity, and the EHR-style structured outputs each chatbot returns will be compared. The result will inform the design of the Stage 4 EHR schema.
2. **Phase 4 execution – LoRA fine-tuning of Qwen3-ASR-1.7B.** The methodology study currently in progress will be operationalized: LoRA adapters will be inserted into the speech encoder, the language-model decoder will be kept frozen, and the adapter will be trained on the multi-dialect corpus. The fine-tuned model will be re-evaluated against the full benchmark.
3. **Corpus expansion.** The audio corpus will be expanded to 10–20 hours, with explicit dialect balancing and the addition of Chittagonian and Rangpuri.
4. **Medical NER model.** A BanglaBERT-based medical NER model will be trained on a small annotated subset of the project's transcripts to populate Stage 3 of the pipeline.
5. **End-to-end integration.** The Stage 4 EHR schema and Stage 5 doctor dashboard will be implemented as a working end-to-end prototype suitable for usability evaluation.
6. **Field evaluation.** A pilot deployment study with a partner clinic, conducted under explicit informed consent and ethics review, will test the system on real patient interactions and report end-to-end accuracy under realistic clinical noise.

References

[1] World Health Organization, "Density of physicians (per 1000 population), bangladesh." https://data.who.int/countries/050, 2023. Online; accessed April 2026.

[2] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," *Proceedings of the 40th International Conference on Machine Learning (ICML)*, pp. 28492–28518, 2023. Available: https://arxiv.org/abs/2212.04356.

[3] V. Pratap, A. Tjandra, B. Shi, P. Tomasello, A. Babu, S. Kundu, A. Elkahky, Z. Ni, A. Vyas, M. Fazel-Zarandi, A. Baevski, Y. Adi, X. Zhang, W.-N. Hsu, A. Conneau, and M. Auli, "Scaling speech technology to 1,000+ languages," *arXiv preprint arXiv:2305.13516*, 2023. Available: https://arxiv.org/abs/2305.13516.

[4] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 12449–12460, 2020. Available: https://arxiv.org/abs/2006.11477.

[5] Seamless Communication, L. Barrault, Y.-A. Chung, M. C. Meglioli, D. Dale, N. Dong, M. Duppenthaler, P.-A. Duquenne, B. Ellis, H. Elsahar, *et al.*, "Seamlessm4t: Massively multilingual and multimodal machine translation," *arXiv preprint arXiv:2308.11596*, 2023. Available: https://arxiv.org/abs/2308.11596.

[6] Y. Chu, J. Xu, Q. Yang, H. Wei, X. Wei, Z. Guo, Y. Leng, Y. Lv, J. He, J. Lin, C. Zhou, and J. Zhou, "Qwen2-audio technical report," *arXiv preprint arXiv:2407.10759*, 2024. Available: https://arxiv.org/abs/2407.10759.

[7] Bengali.AI, "Bengali.ai speech recognition competition." https://www.kaggle.com/competitions/bengaliai-speech, 2023. Online; accessed April 2026.

[8] R. Ardila, M. Branson, K. Davis, M. Henretty, M. Kohler, J. Meyer, R. Morais, L. Saunders, F. M. Tyers, and G. Weber, "Common voice: A massively-multilingual speech corpus," *Proceedings of the 12th Language Resources and Evaluation Conference (LREC)*, pp. 4218–4222, 2020. Available: https://commonvoice.mozilla.org.

[9] AI4Bharat, "IndicWav2Vec: Self-supervised speech models for indian languages." https://github.com/AI4Bharat/IndicWav2Vec, 2022. Online; accessed April 2026.

[10] A. Gupta, H. S. Chadha, P. Shah, N. Chhimwal, A. Dhuriya, R. Gaur, and V. Raghavan, "Vakyansh: ASR toolkit for low resource indic languages." https://github.com/Open-Speech-EkStep/vakyansh-models, 2022. Online; accessed April 2026.

[11] F. R. Rakib, S. S. Dip, S. Alam, N. Tasnim, M. I. H. Shihab, M. N. Ansary, S. M. Hossen, M. H. Meghla, M. Rahman, F. Sadeque, S. S. Chowdhury, T. Reasat, A. Sushmit, and A. I. Humayun, "Ood-speech: A large bengali speech recognition dataset for out-of-distribution benchmarking," in *Proceedings of INTERSPEECH 2023*, 2023. Available: https://arxiv.org/abs/2305.09688.

[12] A. Bhattacharjee, T. Hasan, W. Ahmad, K. S. Mubasshir, M. S. Islam, A. Iqbal, M. S. Rahman, and R. Shahriyar, "Banglabert: Language model pretraining and benchmarks for low-resource language

understanding evaluation in bangla," in *Findings of the Association for Computational Linguistics: NAACL 2022*, pp. 1318–1327, 2022. Available: https://github.com/csebuetnlp/banglabert.

[13] J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C. H. So, and J. Kang, "Biobert: A pre-trained biomedical language representation model for biomedical text mining," *Bioinformatics*, vol. 36, no. 4, pp. 1234–1240, 2020. Available: https://arxiv.org/abs/1901.08746.

[14] E. Alsentzer, J. R. Murphy, W. Boag, W.-H. Weng, D. Jin, T. Naumann, and M. B. A. McDermott, "Publicly available clinical BERT embeddings," in *Proceedings of the 2nd Clinical Natural Language Processing Workshop, NAACL*, pp. 72–78, 2019. Available: https://arxiv.org/abs/1904.03323.

[15] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017. Available: https://arxiv.org/abs/1706.03762.

[16] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of NAACL-HLT 2019*, pp. 4171–4186, 2019. Available: https://arxiv.org/abs/1810.04805.

[17] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz, J. Davison, S. Shleifer, P. von Platen, C. Ma, Y. Jernite, J. Plu, C. Xu, T. L. Scao, S. Gugger, M. Drame, Q. Lhoest, and A. M. Rush, "Transformers: State-of-the-art natural language processing," in *Proceedings of EMNLP 2020: System Demonstrations*, pp. 38–45, 2020. Available: https://github.com/huggingface/transformers.

[18] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *International Conference on Learning Representations (ICLR)*, 2022. Available: https://arxiv.org/abs/2106.09685.

[19] M. Bain, J. Huh, T. Han, and A. Zisserman, "WhisperX: Time-accurate speech transcription of long-form audio," in *Proceedings of INTERSPEECH 2023*, 2023. Available: https://arxiv.org/abs/2303.00747.